

# Calc 1 Answer sheet

September 10, 2026

## Practice

$$\frac{d}{dx} 4x^3 = 12x^2$$

$$\frac{d}{dx} (x^2 + 2x) = 2(x + 1)$$

## More practice

$$1) \frac{d}{dx} 2x^2 + 10 = 4x$$

$$2) \frac{d}{dx} 5x^4 - \frac{2}{3}x^3 = 2x^2(10x - 1)$$

$$3) \frac{d}{dx} 9\sqrt{x} = \frac{9}{2\sqrt{x}}$$

$$4) \frac{d}{dx} \frac{4}{x^2} = -\frac{8}{x^3}$$

$$5) \frac{d}{dx} \frac{2w}{5} = 0$$

## More more practice

$$1) \frac{d}{dx} (10e^x) = 10e^x$$

$$2) \frac{d}{dx} (\ln(x) - \frac{e^2}{3}) = \frac{1}{x}$$

## More more more practice

$$1) \frac{d}{dx} x^3 \cdot x = 4x^3$$

$$2) \frac{d}{dx} e^x \cdot x^2 = e^x x(x + 2)$$

$$3) \frac{d}{dx} (3x^4 - 8)^2 = 24x^3(3x^4 - 8)$$

### Example

For  $x^2y + xy^2 - x$ :

$$\frac{\partial(x^2y + xy^2 - x)}{\partial x} = 2xy + y^2 - 1$$

$$\frac{\partial(x^2y + xy^2 - x)}{\partial y} = x^2 + 2xy$$

### More more more more practice

$$y = \frac{1}{2}x^3 + 3x^2 - 2$$

$$\text{FOC: } y' = \frac{3}{2}x^2 + 6x = 0 \Rightarrow x = 0, x = -4$$

$$\text{SOC: } y'' = 3x + 6$$

- $y''(0) = 6 > 0 \Rightarrow$  local minimum at  $x = 0$  ( $y = -2$ )
- $y''(-4) = -6 < 0 \Rightarrow$  local maximum at  $x = -4$  ( $y = 14$ )

### Practice again

$y = \frac{1}{2}x^3 + 3x^2 - 2$  is a cubic defined on all of  $\mathbb{R}$ :  $y \rightarrow \infty$  as  $x \rightarrow \infty$  and  $y \rightarrow -\infty$  as  $x \rightarrow -\infty$ , so it has no global maximum or minimum – only the local extrema found above.

### Another one

$$f(x) = \frac{x^3}{3} - \frac{3}{2}x^2 - 10x \text{ on } [-6, 6]$$

$$\text{FOC: } f'(x) = x^2 - 3x - 10 = (x - 5)(x + 2) = 0 \Rightarrow x = -2, x = 5$$

Compare critical points and endpoints:

- $f(-6) = -66$
- $f(-2) = \frac{34}{3}$
- $f(5) = -\frac{275}{6}$
- $f(6) = -42$

Global maximum at  $x = -2$  ( $f(-2) = \frac{34}{3}$ ); global minimum at  $x = -6$  ( $f(-6) = -66$ ).\$